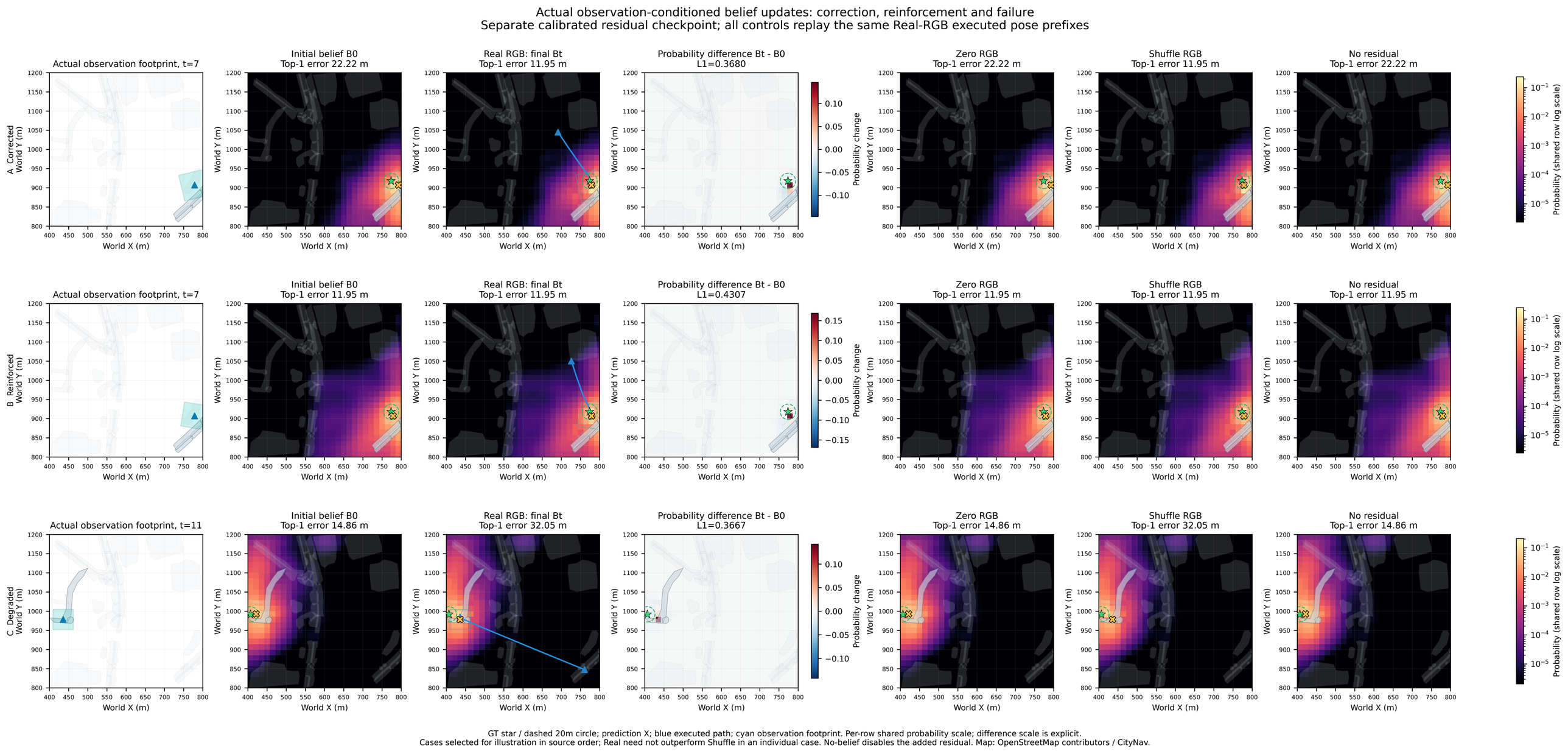

Actual observation-conditioned belief updates: correction, reinforcement and failure
Separate calibrated residual checkpoint; all controls replay the same Real-RGB executed pose prefixes
A. Corrected
B. Reinforced
C. Degraded
Actual observation footprint, t=7
Actual observation footprint, t=7
Actual observation footprint, t=11
Initial belief B0
Top-1 error 22.22 m
Initial belief B0
Top-1 error 11.95 m
Initial belief B0
Top-1 error 14.86 m
Real RGB: final Bt
Top-1 error 11.95 m
Real RGB: final Bt
Top-1 error 11.95 m
Real RGB: final Bt
Top-1 error 32.05 m
Probability difference Bt - B0
L1=0.3680
Probability difference Bt - B0
L1=0.4307
Probability difference Bt - B0
L1=0.3667
Zero RGB
Top-1 error 22.22 m
Zero RGB
Top-1 error 11.95 m
Zero RGB
Top-1 error 14.86 m
Shuffle RGB
Top-1 error 11.95 m
Shuffle RGB
Top-1 error 11.95 m
Shuffle RGB
Top-1 error 32.05 m
No residual
Top-1 error 22.22 m
No residual
Top-1 error 11.95 m
No residual
Top-1 error 14.86 m
World X (m)
World Y (m)
Probability change
Probability (shared row log scale)
GT star / dashed 20m circle; prediction X; blue executed path; cyan observation footprint. Per-row shared probability scale; difference scale is explicit.
Cases selected for illustration in source order; Real need not outperform Shuffle in an individual case. No-belief disables the added residual. Map: OpenStreetMap contributors / CityNav.